LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Eddy current inspection, principle of operation, function of basic system, coil impedance, electrical conductivity mag permeability, types of inspection coils, application of various types of parts
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Eddy current inspection, principle of operation, function of basic system, coil impedance, electrical conductivity mag permeability, types of inspection coils, application of various types of parts
5. Specific Objectives	19.1.1.1 Trainee will be able to describe types of NDT and perform liquid penetrant, magnetic particles and eddy current inspections

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Eddy Current testing principle using non-conductive and conductive samples; Safety precautions regarding high voltage and electrical grounding; Instructor supervision during coil coupling and signal interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) List the three main types of NDT methods covered in this course.

(b) Define the basic principle of electromagnetic induction.

(c) State the relationship between electrical conductivity and eddy current generation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of operation of eddy current inspection • Function of basic system components • Concept of coil impedance and phase angle • Role of electrical conductivity and magnetic permeability • Factors affecting eddy current penetration depth	L & Ds	PPT, IB	15 min
(b)	• Types of inspection coils and their configurations • Application of various coil types on different parts • Interpretation of impedance plane diagrams • Effect of lift-off and frequency on inspection results • Selection criteria for coil type based on defect orientation	L & Ds	PPT, PDF	20 min
(c)	• Standardized testing procedures for eddy current • Calibration and sensitivity settings • Reporting of indications and false calls • Comparison with other NDT methods • Limitations of eddy current inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how magnetic permeability affects the sensitivity of eddy current testing.

(b) Describe the function of the reference standard in calibrating an eddy current system.

(c) Differentiate between absolute and differential probe applications.

12. Home Assignment and Follow Up	Prepare a brief report comparing the applicability of eddy current inspection with radiographic inspection for detecting surface cracks in non-ferromagnetic aircraft structures, referencing the principles covered in this lesson and the upcoming Radiographic inspection lesson (LP 05).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Radiographic inspection, use, applicability inspection principle of radiography, X-ray gamma ray, shadow for motion, types of recording, types of test specimen safety precaution while using x-ray	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min